

Hardware Security & Trust — Homework 4

Question 1 – Watermarking Using Don't-Care Conditions

Given

$$F(a, b, c, d) = \Sigma m(1, 3, 4, 6, 9, 11), D = \Sigma d(0, 5, 15)$$

Tasks

1. Embed the 3-bit author signature “**101**” in F using the don't-care watermarking technique.
 - Update the truth-table entries for minterms 0, 5 and 15.
 2. List the ON-minterms of the water-marked function F_{wm} .
 3. Minimise $F_{wm}(a, b, c, d)$ with Karnaugh maps and give the simplified SOP form.
 4. Explain briefly
 - why normal circuit operation is unchanged, and
 - how the 3-bit signature can later be detected and verified.
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Question 2 – Watermarking “SEC” in a 4-to-2 Encoder

A 4-to-2 encoder is valid only when **exactly one** of the inputs a,b,c,d is high; the other 12 patterns are don't-cares.

Embed the 3-letter string “**SEC**” (8-bit ASCII per letter) into these don't-care patterns using the class method:

1. Write the 24-bit ASCII stream for “SEC”.
2. Starting **from the right-hand end**, split the bit stream into 5-bit groups.
*If the final group is incomplete, append zero(s) to the **right** until the length is a multiple of 5.*
3. Record the new input–output pairs and present the updated truth table.
4. Using 2-variable Karnaugh maps for XX and YY, give the minimal additional product terms and justify that normal encoder operation is unaffected.

Question 3 – Watermarking a Graph-Coloring Solution

Using the edge-insertion rule shown in the lecture slides, embed the **12-bit** binary string to below graph.

101001011111101001011111

